

THE SEAM

Shared Feet and Exiled Meetings

A commentary on Axiom C2 of Double-Slit Geometry — companion note to the Tetra Engine project

Axiom C2 states: *two distinct rays have a common point if and only if they share a foot, and the common point is then that foot, on the seam*. Two phrases in the informal gloss of this axiom — “sharing a foot” and “the meeting is exiled to the seam” — carry the whole mechanism of the geometry. This note unpacks them.

1 • Sharing a foot, precisely

A ray touches the first slit ℓ_1 at exactly one point, its *first foot*, and the second slit ℓ_2 at exactly one, its *second foot* — a transversal that met a slit twice would *be* that slit. Two rays *share a foot* when they pass through the same point of the same slit. Now recall the basic fact about lines in space: two distinct lines meet if and only if they are coplanar. For rays, coplanarity has only one way to happen. If two rays pass through the same point p on ℓ_1 , they obviously meet there; and the plane they span is forced to contain ℓ_2 , because each ray hands that plane one point of ℓ_2 and two points pin the line. That plane is $\text{span}(p, \ell_2)$, and the two rays belong to the flat pencil at p . Conversely, if two rays have distinct feet on *both* slits, they cannot meet anywhere: a meeting point off the slits would be a scene point threaded by two rays, contradicting the Threading Theorem, and a meeting point *on* a slit would be a shared foot by definition.

So “sharing a foot” is simultaneously three things: **co-pencillary**, **coplanar**, and **incident**. One small fact closes the case: two distinct rays can share at most one foot, since sharing both would make them the same line. The two parallelisms of the geometry never overlap.

2 • Where the meetings went

Meetings are not abolished — a camera with no incidences at all could not have pencils, and pencils are its planes. They are *sorted and relocated*. Count concurrence by how many rays pass through a point:

Location	Rays through it	Character
a pinhole center	∞^2 — a full bundle	every ray, all at once, at one interior point
a seam point	∞^1 — a pencil	a half-pinhole

a generic scene exactly 1
point

meeting-free interior

The pinhole spends its entire two-parameter ray budget at a single point. The double slit spends one dimension of concurrence along each slit: every point of the vertical slit is where one image-*column*'s rays cross; every point of the horizontal slit is where one image-*row*'s rays cross. The meetings have been factorized by coordinate — x-meetings deported to one line, y-meetings to the other, at two different depths. That factorization is anamorphic dissociation. And it is why depth becomes measurable: the pinhole organizes the image completely, once, at one depth, leaving nothing to compare; the double slit organizes it twice, partially, at Z_v and Z_h , and the mismatch between the two partial organizations is exactly what the Aperture Cross-Ratio Theorem reads off as the aspect ratio.

3 • Exile, and the meeting-free interior

“Exiled” also means the scene interior is meeting-free. Off the seam, no two rays ever touch — that is what makes (S, T, U) a coordinate system: no scene point owns two addresses. The addressing fails only *on* the seam, and it fails in a telling way: a seam point p on ℓ_1 lies on a whole pencil, so its S is determined but its T is free — the address does not vanish, it blows up. Literally: blowing up the two slit lines resolves the singularity, unfolding each seam point into the projective line of rays through it. The point recovers a unique address if one remembers the direction of approach — the excluded seam, found after the fact.

4 • The inversion in the image

Here is the part most worth staring at. A chain — the image of a straight line — crosses each pencil exactly once, so no chain can contain two rays that share a foot. Meanwhile any two rays with distinct feet on both slits are joined by a one-parameter family of chains. Therefore: **two rays meet in space if and only if no line's image joins them in the picture**. Incidence in the world becomes prohibition in the image. This is precisely the “parallelism” of a Minkowski plane: parallel points are the unjoinable ones. Exile is not mere relocation — the meeting relation crosses over and reappears on the image side wearing the opposite sign.

5 • The algebra of the seam

The coordinatizing number system says the same thing arithmetically. In the double numbers $\mathbb{R} \oplus \mathbb{R}$, two ray-addresses are joinable when their difference is a unit, parallel when it is a zero divisor — a pair of the form (a, 0) or (0, b). The zero divisors *are* the seam: meetings live exactly where division fails. This is why the split algebra appears rather than $\mathbb{C}$: the complex numbers have no zero divisors, so

Möbius geometry has no parallel points and no seam — the pinhole-family analogue. And since $\mathbb{R} \oplus \mathbb{R}$ is the split-complex plane of 1+1 relativity, the zero divisors are its light cone: two rays that meet in space are *null-separated* in the image. The seam is the picture's light cone.

6 • A capstone, hedged

One classification result frames the whole discussion, recalled here from classical line geometry and flagged as such: Kummer's study of congruences of order one says that if every scene point is to be threaded by exactly one ray, the aperture has very few options — a point (the pinhole), two skew lines together with their degenerations (this camera, and the pushbroom limit), or the secants of a twisted cubic. The exile constructed here is not one arbitrary choice among many; it is nearly the only alternative geometry permits to the pinhole.

Companion to Double-Slit Geometry: Definitions, Axioms, and Theorems. Terms used here — ray, foot, pencil, chain, seam, Threading Theorem, Aperture Cross-Ratio Theorem, Axiom C2 — are defined there; numerical verification of the underlying claims is in `geometry_checks.py`.